

Benefit-Cost Ratio Calculator Documentation

1 Introduction

The `bcr_calculator.py` script calculates Benefit-Cost Ratios (BCR) and Net Benefits for transmission projects. It aggregates benefits and costs from all CTCC modules and computes multiple BCR metrics to evaluate project economics from different perspectives.

2 Method Overview

2.1 Purpose

This script calculates BCR metrics by:

1. Loading scenario data from `batch_summary.csv` (robust lookup: exact, partial, prefix, last-row fallback)
2. Aggregating benefits (congestion, curtailment; line losses are always costs, never benefits)
3. Aggregating costs by category (capital, operational, energy/emissions, risk, delay) using `safe_get_numeric` and `prefer_csv_subtotal`
4. Applying optional `BCRConfig` exclusions (`no_emissions`, `no_linelosses`, `capital_only`, `no_wildfire`, etc.) when loading from YAML
5. Calculating multiple BCR metrics (system, capital, excluding risk, excluding emissions)
6. Computing net benefits; displaying formatted summary and updating CSV

2.2 Calculation Approach

The BCR calculation follows this structure:

$$\text{Total Benefits (Conservative)} = \text{Congestion (Haircut)} + \text{Curtailment (Haircut)} \quad (\text{line losses are costs, not}) \quad (1)$$

$$\text{Total Costs} = \text{Capital} + \text{Operational} + \text{Energy/Emissions} + \text{Risk} + \text{Delay} \quad (2)$$

$$\text{BCR}_{\text{system}} = \frac{\text{Conservative Benefits}}{\text{Total Costs}} \quad (3)$$

$$\text{BCR}_{\text{capital}} = \frac{\text{Conservative Benefits}}{\text{Capital Costs}} \quad (4)$$

$$\text{BCR}_{\text{excluding_risk}} = \frac{\text{Conservative Benefits}}{\text{Total Costs} - \text{Risk Costs}} \quad (5)$$

$$\text{BCR}_{\text{excluding_emissions}} = \frac{\text{Conservative Benefits}}{\text{Total Costs} - \text{Emissions Costs}} \quad (6)$$

$$\text{Net Benefit} = \text{Conservative Benefits} - \text{Total Costs} \quad (7)$$

2.3 Inputs and Outputs

Inputs:

- Scenario ID (to load from `batch_summary.csv`)
- All cost and benefit metrics from CTCC modules

Outputs:

- BCR metrics (system, capital, excluding risk, excluding emissions) - all use conservative benefits
- Net benefits (total and excluding risk)
- Benefit and cost breakdowns
- Formatted summary display

2.4 Integration with Other Scripts

This script integrates with:

- `path_config.py` — `OUTPUTS_DIR`, `YAMLS_DIR` (BCR config YAML optional)
- `batch_summary.csv` — Reads scenario row by `scenario_id`; writes BCR columns back
- All CTCC cost/benefit scripts — Aggregates their outputs from the batch summary row

3 Key Functions

3.1 `load_scenario_data(scenario_id, output_dir)`

Loads scenario data from `batch_summary.csv`. Uses robust lookup: (1) exact match on `scenario_id`, (2) partial match (substring), (3) prefix match (e.g. `sample_0_...`), (4) fall-back to last row (for batch sensitivity runs). Converts numeric cells via `convert_row_to_numeric`.

Parameters:

- `scenario_id` (str): Scenario identifier
- `output_dir` (str): Directory containing `batch_summary.csv` (default `OUTPUTS_DIR`)

Returns:

- Dictionary with all data for the scenario, or `None` if file missing/empty

3.2 `calculate_benefits(data)`

Calculates total benefits from scenario data.

Parameters:

- `data` (dict): Dictionary with scenario data

Returns:

- Dictionary with benefit breakdown and totals (PV, nominal, haircut)

Key Logic:

```
1 congestion_benefit_pv = safe_get_numeric(data, "  
    congestion_benefit_pv")  
2 curtailment_benefit_pv = safe_get_numeric(data, "  
    curtailment_benefit_pv")  
3 # Line losses are always costs, never benefits (line_loss_benefit_pv  
    = 0)  
4 calculated_total_benefits = congestion_benefit_pv +  
    curtailment_benefit_pv  
5 total_benefits_pv = prefer_csv_subtotal(data, "total_benefits_pv",  
    calculated_total_benefits)  
6 congestion_benefit_haircut = safe_get_numeric(data, "  
    congestion_benefit_haircut_pv")  
7 curtailment_benefit_haircut = safe_get_numeric(data, "  
    curtailment_benefit_haircut_pv")  
8 total_benefits_haircut_pv = prefer_csv_subtotal(data, "  
    total_benefits_haircut_pv",  
9    congestion_benefit_haircut + curtailment_benefit_haircut)
```

3.3 calculate_costs(data)

Calculates total costs from scenario data, organized by category.

Parameters:

- **data** (dict): Dictionary with scenario data

Returns:

- Dictionary with cost breakdown by category and totals (PV and nominal)

Cost Categories:

- **Capital:** Build, ROW, environmental mitigation
- **Operational:** O&M, insurance, line losses (greenfield), emissions
- **Risk:** Wildfire, outage
- **Delay:** Construction delay, congestion/curtailment delay, residual congestion

Key Logic:

```
1 # Capital costs
2 capital_costs_pv = build_cost_pv + row_cost_pv + env_mitigation_pv
3
4 # Operational costs (line losses only if positive - greenfield)
5 line_loss_cost_pv = max(0, line_loss_pv)
6 operational_costs_pv = oandm_pv + insurance_pv + line_loss_cost_pv +
   emissions_pv
7
8 # Risk costs
9 risk_costs_pv = wildfire_pv + outage_pv
10
11 # Delay costs
12 delay_costs_pv = (delay_cost_pv + congestion_delay_pv
13                  + curtailment_delay_pv + residual_congestion_pv)
14
15 # Totals
16 total_costs_pv = (capital_costs_pv + operational_costs_pv
17                  + risk_costs_pv + delay_costs_pv)
```

3.4 calculate_bcr_metrics(benefits, costs)

Calculates benefit-cost ratios and net benefits.

Parameters:

- **benefits** (dict): Dictionary with benefit breakdown
- **costs** (dict): Dictionary with cost breakdown

Returns:

- Dictionary with BCR metrics and net benefits

BCR Metrics:

- **BCR_system**: Conservative benefits / Total costs (primary metric, uses saturation-adjusted benefits)
- **BCR_capital**: Conservative benefits / Capital costs (investor perspective, uses saturation-adjusted benefits)
- **BCR_excluding_risk**: Conservative benefits / (Total costs - Risk costs) (core economics, uses saturation-adjusted benefits)
- **BCR_excluding_emissions**: Conservative benefits / (Total costs - Emissions costs) (uses saturation-adjusted benefits)

Key Logic:

```
1 # Calculate costs excluding risk
2 total_costs_excluding_risk_pv = total_costs_pv - risk_costs_pv
3
4 # Use conservative (haircut) benefits for all BCR calculations
5 total_benefits_pv = total_benefits_haircut_pv
6
7 # Calculate BCRs (with division by zero protection)
8 bcr_system = total_benefits_pv / total_costs_pv if total_costs_pv >
   0 else 0
9 bcr_capital = total_benefits_pv / capital_costs_pv if
   capital_costs_pv > 0 else 0
10 bcr_excluding_risk = (total_benefits_pv /
   total_costs_excluding_risk_pv
11                       if total_costs_excluding_risk_pv > 0 else 0)
12 bcr_excluding_emissions = (total_benefits_pv /
   total_costs_excluding_emissions_pv
13                           if total_costs_excluding_emissions_pv > 0
   else 0)
14
15 # Net benefits (using conservative benefits)
16 net_benefit_pv = total_benefits_pv - total_costs_pv
17 net_benefit_excluding_risk_pv = total_benefits_pv -
   total_costs_excluding_risk_pv
```

3.5 calculate_and_display_bcr(scenario_id, output_dir)

Main function to calculate and display BCR analysis.

Parameters:

- `scenario_id` (str): Unique identifier for the scenario
- `output_dir` (str): Directory containing `batch_summary.csv`

Returns:

- Dictionary with all BCR results, or None if calculation fails

3.6 `print_bcr_summary(benefits, costs, bcr_metrics, data)`

Prints formatted BCR summary to terminal.

4 Example

The following example demonstrates a typical usage:

```

1 # Example: 500 MW Overhead AC line
2 # Scenario ID: "20250101_120000"
3 #
4 # Inputs (from batch_summary.csv):
5 #   - Congestion benefit PV: $500M
6 #   - Curtailment benefit PV: $200M
7 #   - Line loss cost PV: $10M (greenfield - cost)
8 #   - Build cost PV: $100M
9 #   - ROW cost PV: $20M
10 #   - Environmental PV: $15M
11 #   - O&M PV: $50M
12 #   - Insurance PV: $5M
13 #   - Emissions PV: $8M
14 #   - Wildfire PV: $1B
15 #   - Outage PV: $500M
16 #   - Delay costs PV: $2M
17 #   - Residual congestion PV: $50M
18 #
19 # Calculation:
20 #   Total Benefits = $500M + $200M + $0 = $700M
21 #   Capital Costs = $100M + $20M + $15M = $135M
22 #   Operational Costs = $50M + $5M + $10M + $8M = $73M
23 #   Risk Costs = $1B + $500M = $1.5B
24 #   Delay Costs = $2M + $50M = $52M
25 #   Total Costs = $135M + $73M + $1.5B + $52M = $1.76B
26 #
27 #   BCR_system = $700M / $1.76B = 0.40
28 #   BCR_capital = $700M / $135M = 5.19
29 #   BCR_excluding_risk = $700M / ($1.76B - $1.5B) = 2.69
30 #   Net Benefit = $700M - $1.76B = -$1.06B
31 #
32 # Output:

```

```

33 # - BCR_system: 0.40
34 # - BCR_capital: 5.19
35 # - BCR_excluding_risk: 2.69
36 # - Net benefit: -$1.06B

```

Integration Example:

```

1 # When called from ctcc.py:
2 from scripts.bcr_calculator import calculate_and_display_bcr
3
4 # After all other scripts complete
5 scenario_id = os.environ.get('CTCC_SCENARIO_ID')
6 results = calculate_and_display_bcr(scenario_id)
7
8 # Update batch_summary.csv with BCR metrics
9 csv_manager = CTCCOutputManager()
10 csv_manager.add_bcr_metrics(results)
11 csv_manager.write_batch_summary()

```

5 Key Implementation Details

5.1 Line Loss Treatment

The script handles line losses differently for greenfield vs. reconductoring:

- **Greenfield:** Line losses are positive (costs) - counted in operational costs
- **Reconductoring:** Line losses are negative (benefits) - counted in benefits

5.2 Multiple BCR Perspectives

The script calculates multiple BCR metrics to provide different perspectives. All BCR metrics use conservative (saturation-adjusted) benefits to account for diminishing marginal returns:

- **System BCR:** Overall project economics including all costs (uses conservative benefits)
- **Capital BCR:** Benefits relative to capital investment (investor view, uses conservative benefits)
- **Excluding Risk BCR:** Core economics without wildfire/outage risk costs (uses conservative benefits)
- **Excluding Emissions BCR:** Economics without emissions costs (uses conservative benefits)

5.3 Net Benefits

The script calculates net benefits in addition to BCRs:

- **Net Benefit:** Total benefits - Total costs
- **Net Benefit Excluding Risk:** Total benefits - (Total costs - Risk costs)

5.4 Division by Zero Protection

All BCR calculations include checks to prevent division by zero, returning 0 if the denominator is zero or negative.

5.5 Data Loading

The script loads data from `batch_summary.csv` using the scenario ID, automatically converting numeric strings to floats for calculations.